



ARTICLES AND BRIEFINGS | Trending Issues In IP Litigation

As The Pendulum Swings: Re-Exploring Nuclear Energy and The State of The Art

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Increasing demand for reliable, baseload power and concerns related to climate change are motivating a transition toward new energy sources and alternatives to fossil fuels. Nuclear energy currently provides approximately 20% of US electricity without releasing greenhouse gases and other pollutants into the air. While radioactive waste as a by-product is unavoidable and requires careful storage and management, nuclear energy's ample ability to provide baseload power makes it an intriguing and relevant energy alternative. Of course, advances in technology go hand-in-hand with litigation meant to protect those advances, and nuclear power is not without its legal battles. Yet, if the most recent example of nuclear competitors failing to get along starts a trend, litigation in the nuclear space may actually spark more collaboration between competitors than confrontation.

ARE WE IN A NUCLEAR RENAISSANCE?

Since the first grid-connected nuclear powerplant in Russia in 1954, [32 countries](#) around the world have constructed 440 nuclear reactors. The other 160+ countries are in various stages of nuclear development: some, like Australia, have banned nuclear power usage altogether; others are either phasing out, or have phased out, nuclear power; and still others are reconsidering or actively reviving nuclear power. While nuclear power has never gone away, in this context of mixed stages of nuclear development, the energy alternative is receiving more attention.

In the United States in particular, there appears to be a nuclear resurgence under the Trump Administration. While President Trump has been in office, the White House has [actively encouraged](#) nuclear research, and government agencies have highlighted advancements accomplished within this energy sector during the administration's [first 100 days](#). Such advancement has included strengthening the nuclear fuel supply chain by reducing U.S. dependence on China-sourced cobalt, a key element used in nuclear reactors. The administration also supported the nation's first restart of a retired nuclear power plant (Palisades Nuclear Plant in Michigan). Moreover, the administration continues to encourage nuclear testing in labs around the U.S. And in relation to oversight and administrative control of this boom in nuclear research, the president signed an [executive order](#) that would reorganize the Nuclear Regulatory Commission and overhaul its regulations with the intention of quicker reviews of nuclear projects.

Outside of the political realm, the worldwide surge in AI is also providing an opportunity for nuclear power to take center stage. Data centers, a vital requirement for running AI systems, need power, massive amounts of power, and there has been a flurry of activity in the nuclear space as tech giants look [for energy sources to power these data centers](#).

In this context, there is renewed interest in nuclear energy.

SPLITTING THE ATOM: STATE OF THE ART

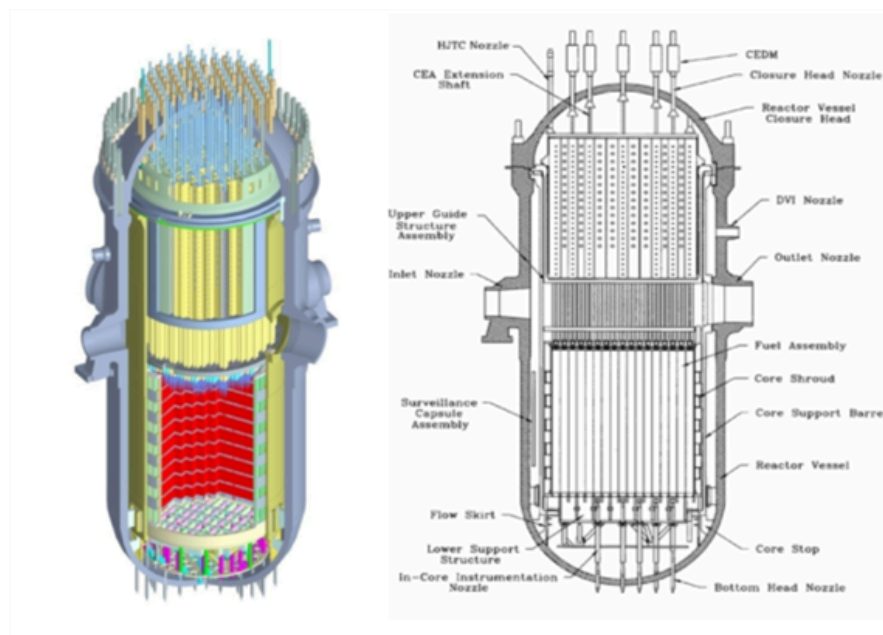
Recent technological advances in nuclear reactor designs address many of the longstanding worries about nuclear energy. For example, Generation III+ and IV reactors incorporate [passive safety features](#) and improvements based on decades of operational experience. [Molten salt reactors](#) provide an alternative to water-cooled systems, and [thorium-based fission](#) provides a uranium-adjacent alternative that uses a more abundant resource and, in some respects, may be superior in physical and nuclear fuel properties. Additionally, smaller scale reactors, called “small modular reactors,” and even microreactors, are in development. A smaller footprint offers more flexible placement options, cost-effective upkeep and construction, faster construction, and scalability.

The nuclear power technology the United States and other countries have relied upon for decades continues to see innovation.

NUCLEAR INNOVATION BREEDS LITIGATION. . . AND COLLABORATION?

With innovation in any technology comes litigation focused on protecting that innovation. One particular case in the nuclear space provides an informative view into international nuclear energy relations and nuclear intellectual property litigation that melds rights of IP owners with overseas competitors, and the resulting impact on the global nuclear market. This particular case also highlights how, even if a party loses the adversarial battle in court, concurrent arbitration may actually win them the IP rights war by encouraging collaboration with competitors to reach agreements that benefit all parties involved.

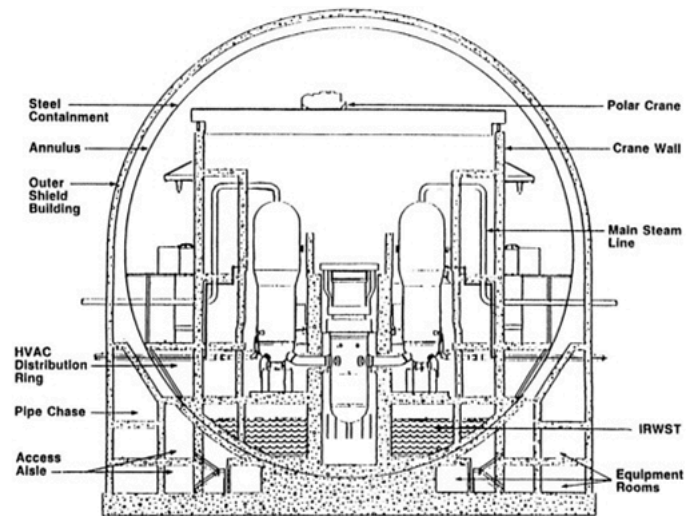
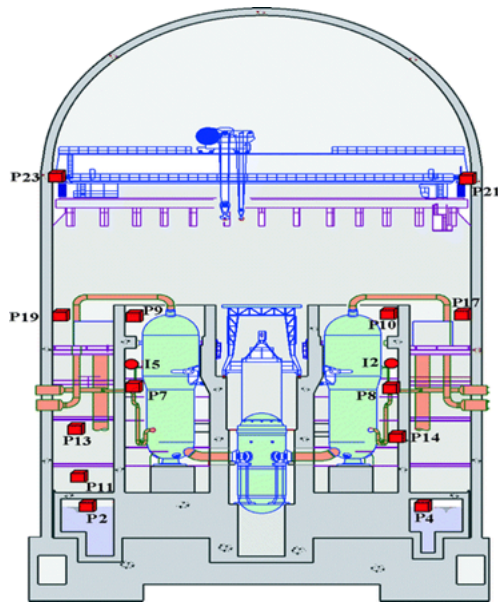
In October 2022, Westinghouse Electric Company, a U.S.-based major nuclear player, filed *Westinghouse Electric Company LLC v. Korea Electric Power Corporation (KEPCO) and Korea Hydro & Nuclear Power Co. Ltd. (KHNP)* (Case No. 1:22-cv-03228-APM), in D.C. District Court. Westinghouse alleged that KEPCO and KHNP’s APR1400 (Advanced Power Reactor 1400) technology copied the System 80 reactor designed by Combustion Engineering, which Westinghouse acquired in 2000. Westinghouse filed the lawsuit to stop the Korean entities from selling reactors to Poland, which was in the process of evaluating offers from various nuclear power companies, Westinghouse included, to supply the country’s first commercial nuclear powerplant. Saudi Arabia and the Czech Republic had also invited KEPCO and KHNP to make bids for nuclear powerplants.



Reactor vessel of the APR1400

The APR1400 is a pressurized water reactor with its origins in the Combustion Engineering System 80+ design. Principally designed by KEPCO, the APR1400 produces 1400 MWe and has a 60-year design life. The APR1400 features improvements in operation, safety, maintenance, and affordability. In comparison to some of its peers, the APR1400 is [faster to construct than the](#)

AP1000 has a design life on the longer end of the spectrum. It is a design that is promising to those interested in investing in nuclear power.



APR1400 vs. System 80+

In the lawsuit, Westinghouse specifically alleged that the use of APR1400 technology required its permission before being transferred to other countries considering deploying the design. This is because, Westinghouse claimed, U.S. laws restricting nuclear power sharing—known as Assistance to Foreign Atomic Energy Activities, 10 CFR Part 810 (Part 810)—require the technology owner to grant permission for use. Part 810 sets out regulations governing technology transfer related to nuclear reactors, equipment, and materials, and also requires the U.S. Department of Energy approve that sharing. Its purpose is stated in § 810.1:

“The regulations in this part implement section 57 b.(2) of the Atomic Energy Act, which empowers the Secretary, with the concurrence of the Department of State, and after consultation with the Nuclear Regulatory Commission, the Department of Commerce, and the Department of Defense, ***to authorize persons to directly or indirectly engage or participate in the development or production of special nuclear material outside the United States.***”

Westinghouse interpreted the requirements in Part 810 to allow the owner of the technology at issue to weigh into its shipping overseas as well.

KEPCO and KHNP responded by filing countersuits against Westinghouse, and a motion to dismiss, arguing that Westinghouse’s position—that it had a private right of action to enforce Part 810 under the Atomic Energy Act (“AEA”)—was incorrect because, in the Defendants’ interpretation, the AEA shifts enforcement authority to the U.S. Attorney General, not private entities, such as Westinghouse, as a means to claim rights through litigation.

Ultimately, the District Court agreed with KEPCO and KHNP and dismissed the case in 2023, holding that Westinghouse lacked a private cause of action to enforce Part 810 under the AEA because the AEA vests in the Attorney General the power to seek to enjoin both actual and imminent violations of the Act or its implementing regulations.

So why is this case still relevant in 2025, and where does the collaboration between competitors appear? While Westinghouse may have been unable to state a cause of action under Part 810 to enforce its IP rights, concurrent arbitration, that just reached settlement in January 2025, provided an opportunity for the parties to negotiate. In January, when Westinghouse announced a global settlement agreement with KEPCO and KHNP, the terms of the agreement were still unknown. But this did not dampen what could be classified as an exultant mood with Westinghouse stating that the agreement “sets the stage for future cooperation between the parties to advance new nuclear projects globally.” Note the use of the word “globally”; this agreement appears to affect more than just competing bids for providing Poland with nuclear power. Westinghouse and KEPCO/KHNP focused on new nuclear projects across the globe when they reached their settlement, and large organizations, working together to advance nuclear energy plant construction and production will inevitably cover more ground than they could alone and embroiled in litigation.

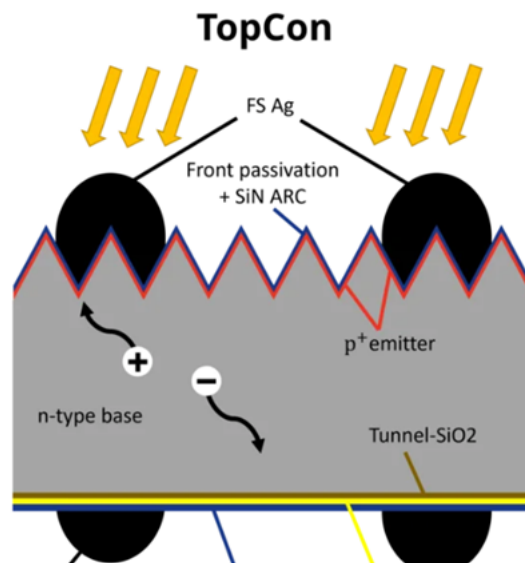
Of the agreement, Tim Gitzel, president and CEO of Cameco Corporation, which owns Westinghouse, [stated](#): “With more than 30 countries and over 100 companies pledging to triple nuclear capacity by 2050, the demand for nuclear power is undeniable. This agreement strengthens the industry’s ability to provide carbon-free, reliable, dispatchable baseload electricity to help achieve climate, energy and national security objectives.” Again, Gitzel undeniably referred to expanding nuclear power across the globe. So, while initially contentious, the APR1400-centered dispute between Westinghouse and KEPCO/KHNP may have positive and exciting ramifications for nuclear energy both at home and abroad.

Perhaps a surprise to see major competitors working together, more collaboration between intellectual-property-owning entities in the nuclear energy space can only encourage growth. And with the number of nuclear power supporters seeming to grow every day, an accessible and litigation-less nuclear power market could result in faster and more efficient growth in the nuclear energy sector.

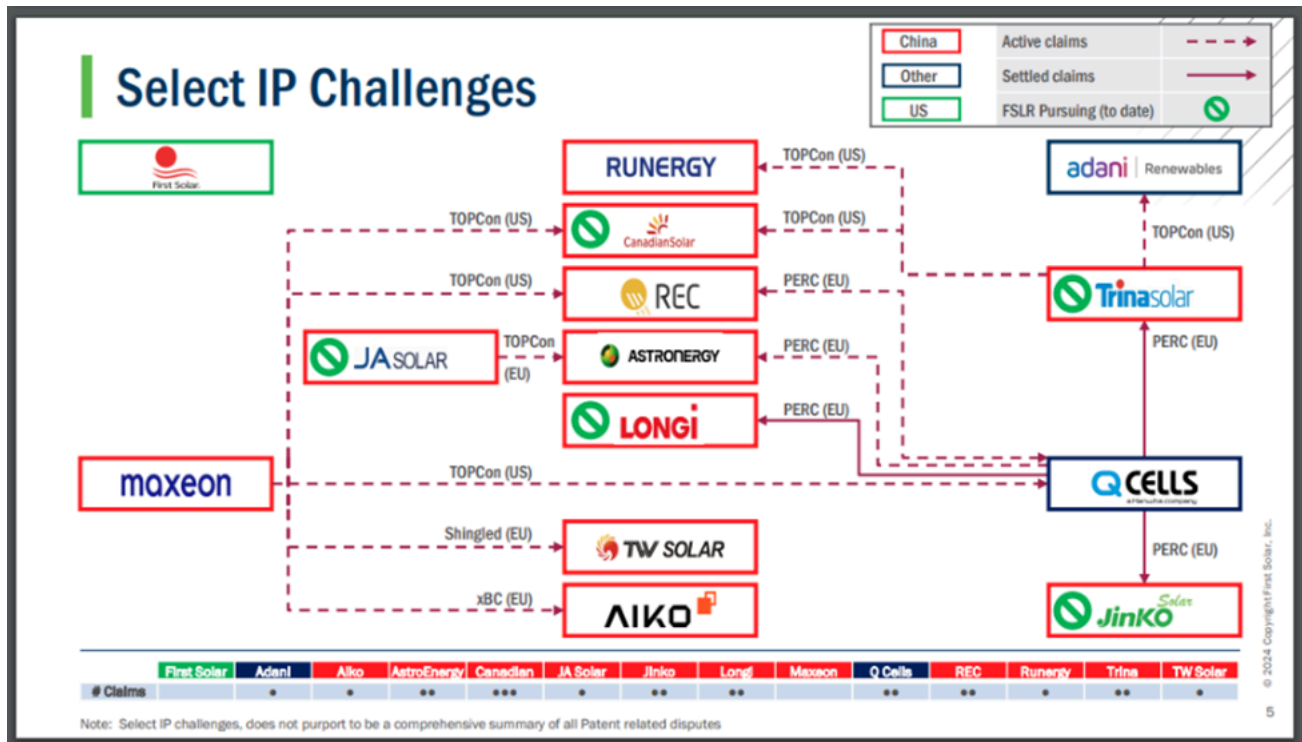
ON THE OTHER HAND: SOLAR INDUSTRY SEES RISE IN IP LITIGATION

While the nuclear energy space appears to have a boom in its near future with more opportunities for collaboration, what about other energy sectors? Has there been recent collaboration in, say, the solar energy space? In short, no.

After decades of research and development, PERC (passivated emitter rear contact) technology has become the dominant market choice in the solar energy sector. Improvements to PERC’s central design usually go one of two ways: either HJT (heterojunction technology) or TOPCon (tunnel oxide passivated contact). Most of the solar industry has decided to take the TOPCon route, which is easier to achieve as it only requires adding one oxide layer to already existing PERC cells. In contrast, HJT’s more complex formula adds a thin, amorphous silicon film layer to both the front and back of solar wafers.



With a proliferation of TOPCon-based panel manufacturers, a crowded market continue to fuel intellectual property disputes. Unlike the collaboration that appears to be taking place in the nuclear space, there is currently a fight for solar supremacy, as he who owns the rights to prevalent solar panel techniques, owns the domestic manufacturing landscape.

Current [chart](#) of TOPCon and TOPCon adjacent litigation

Many of these solar lawsuits are in their nascent stages and only time will tell who the winners and losers of the TOPCon wars will be. First Solar, Inc., an American solar panel manufacturer, appears to be one of the most aggressive players, and the chart above comes from a recent First Solar earnings report, tracking its own litigation and that of its competitors.

WHAT'S NEXT?

As solar companies fight over IP, will we see more nuclear—or other clean energy alternatives—take solar's place? Will there be more collaboration with their respective technology like Westinghouse, KEPCO, and KHNP? Will, and should, the Trump Administration's support for nuclear power research and testing usher in a new era of nuclear power in the U.S.? And what antitrust concerns, if any, arise from large entities working together (Westinghouse, KEPCO, KHNP), or one company reigning supreme in a current energy space (potentially First Solar)? These questions, and more, remain unanswered, at least for now. But there is no denying the growing interest in nuclear power, and how it differs from other energy sectors.

Womble attorneys have extensive experience in the energy sector and are positioned to guide clients through these complex developments; the firm even has an [ongoing series on the subject of nuclear energy](#) we encourage you to read through. Do not hesitate to reach out to your usual Womble counsel with any questions, comments, or concerns.

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